Maximilian Harkins

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Education

University of Rochester — Rochester, NY
B.S. Biomedical Engineering; GPA: 3.71/4.00; Schwartz Discover Grant — *Expected May 2027*

Engineering Experience

Carnegie Mellon University × Meta Reality Labs — Pittsburgh, PA
NeuroMechatronics Lab | Researcher — *May–Aug. 2026*

- Evaluated wearable EMG for computer access after cervical spinal cord injury using hybrid trackball pointing, gesture selection, 2D target acquisition, Fitts-law throughput, and click latency.
- Piloted a forearm IMU pointer with EMG clicking and quaternion-based cursor projection; developed an anatomical framework to interpret sparse forearm EMG for subject-specific gesture decoding.

University of Rochester, Brown Lab — Rochester, NY
Undergraduate Researcher | Advisor: Edward B. Brown — *Dec. 2023–Present*

- Built WayPoint, a MATLAB 3D laser micromachining platform integrating microscope control, CAD/G-code path planning, scanning, and automated experiment logging.
- Designed a motorized XY stage and image/CAD-to-laser trajectory algorithms; characterize collagen ablation with second-harmonic imaging and perfusion assays.

University of Rochester Medical Center, Zhang Lab — Rochester, NY
Undergraduate Researcher | Advisor: Xinping Zhang — *Sept. 2023–Present*

- Converted in vivo mouse-bone microvascular images into 3D laser paths and used them to ablate vessel-derived patterns in collagen.
- Designed microfluidic collagen chambers and casting protocols; used fluorescent-microsphere assays to test channel connectivity and observe preliminary perfusion.

Selected Projects

Two-Photon Polymerization Microfabrication — *Sept. 2026–Present*

- Adapted WayPoint's multiphoton laser-scanning platform from subtractive laser ablation to additive 3D microfabrication using low-cost commercial UV-curable resin.
- Optimized laser power and scan speed for localized photopolymerization, achieving sub-micron lateral and axial fabrication resolution.
- Designing and fabricating fully integrated microscale optical systems with custom 3D lens geometries and arbitrary optical architectures.

Wearable BLE Protocol and Cross-Platform Client — *2026*

- Reverse-engineered a commercial EMG wristband protocol with BLE/LE CoC transport, encrypted sessions, enrollment, RPCs, and EMG/IMU telemetry; built macOS, iOS, and Android clients.

Sparse EMG Decoding and Motor-Unit Decomposition — *Aug.–Sept. 2026*

- Built eight-channel EMG pipelines for 22-joint hand-pose regression and custom gesture classification with separate contraction-intensity estimation using spatial-spectral features and PyTorch.
- Developed offline and online motor-unit decomposition to identify units, recover spike trains, and estimate firing rates for gesture recognition and proportional control.

Presentations

Lead Author, Society for Neuroscience 2026 — Hybrid trackball point-and-click using EMG gestures after cervical spinal cord injury.
Invited Presenter, Hajim National Council 2024 — Multiphoton ablation and laser-guided angiogenesis.

Leadership

Vice President, Quantitative Finance Club — *Feb. 2026–Present*
Research Ambassador, Undergraduate Research — *Apr. 2025–Present*

Technical Skills

Programming: MATLAB, Python, C/C++, Kotlin Multiplatform
Software: PyTorch, MediaPipe, Solidworks, Digilent WaveForms
Hardware/Lab: EMG wearables, BLE/RF, two-photon microscopy, logic analyzers, microfluidics